

Q1. Write a Python program to create a customer_DataFrame with suitable attributes such as Customer_ID, Gender, City, and Membership_Type. Perform label encoding technique on appropriate categorical attributes. Display the DataFrame before and after encoding.

[10 Marks]

Q2. Generate a DataFrame containing customer information for Netflix and Amazon Prime users. Create a Venn diagram and a Pie chart to show the customers using only Netflix, only Amazon Prime, and both services. Add appropriate labels, colors, and titles to both visualizations.

[20 Marks]

OR

Q2. Use the K-Nearest Neighbors (KNN) algorithm to classify emails as spam or non-spam using the Spambase dataset. Preprocess the data, train the KNN classifier, and evaluate its performance using accuracy, confusion matrix, precision, recall, and F1-score.

[20 Marks]

Q3. Viva

[5 Marks]